

DRY-RUN GUIDE | insertion.py
Source: Dsa\sort\insertion.py

CORE BEHAVIOR

Orders values using the comparisons, swaps, insertion, partition, or merges shown in this file.

REPRESENTATIVE INPUT

[4,1,3,2].

STEP TRACE

Each pass/partition places values nearer their final positions; completed sequence is nondecreasing.

EXPECTED OUTPUT / RETURN

1 2 3 4 (check source for exact formatting).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\sort\insertion.py; line numbers refer to the original file. No source was modified.

```
001 | def insertion_sort(arr):  
002 |     for i in range(1, len(arr)):  
003 |         key = arr[i]  
004 |         j = i - 1  
005 |         while j >= 0 and arr[j] > key:  
006 |             arr[j + 1] = arr[j]  
007 |             j -= 1  
008 |         arr[j + 1] = key  
009 |     return arr
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.